Lecture 13

Adding angular momenta

Objectives. *Two spins rotate together. The total angular momentum of a pair is the sum of the two, and it is again an angular momentum in the sense of Lecture 7. So the pair splits into multiplets, and the counting of Lecture 7 tells which. Two multiplets of spins j_1 and j_2 give one multiplet of each spin from $|j_1 - j_2|$ to $j_1 + j_2$. Two spins one half make spin 1, the triplet, and spin 0, the singlet ψ^- of hydrogen's ground state. For $a \neq 0$, an interaction of the form $a\,\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)}$ has one level for each total spin. Inside one multiplet every vector operator acts as a multiple of $\mathbf{J}$, and this gives the splitting of an atomic level in a weak magnetic field.*

By the end of this lecture you will be able to:

- Find the total spins of two multiplets, and check them by counting dimensions.
- Write the coupled basis of two spins one half, and of spin one and spin one half, in terms of products.
- Find the levels of a coupling $a\,\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)}$ with $a \neq 0$, and their degeneracies.
- Find the Landé factor of a level of spin j, and the level's first-order splitting in a weak magnetic field.

13.1 The total angular momentum

Let $\mathbf{J}^{(1)}$ be an angular momentum on $\mathbb{C}^{n_1}$ and $\mathbf{J}^{(2)}$ an angular momentum on $\mathbb{C}^{n_2}$ (Definition 7.1). On the pair $\mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2}$ put

$$J_a = J_a^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes J_a^{(2)}, \qquad a = x, y, z. \tag{13.1}$$

Lemma 13.1 (The total angular momentum)**.** *The matrices J_x, J_y, J_z of (13.1) form an angular momentum on $\mathbb{C}^{n_1} \otimes \mathbb{C}^{n_2}$. For every unit vector $\mathbf{n}$ and every angle θ,*

$$e^{-\mathrm{i}\theta\,\mathbf{n}\cdot\mathbf{J}/\hbar} = e^{-\mathrm{i}\theta\,\mathbf{n}\cdot\mathbf{J}^{(1)}/\hbar} \otimes e^{-\mathrm{i}\theta\,\mathbf{n}\cdot\mathbf{J}^{(2)}/\hbar}.$$

Proof. Each J_a is self-adjoint, since $(A \otimes \mathbb{1})^\dagger = A^\dagger \otimes \mathbb{1}$ (*Linear Algebra*, Prop. 12.9). A matrix $A \otimes \mathbb{1}$ commutes with every $\mathbb{1} \otimes B$. So in $\big[J_x, J_y\big]$ the two cross terms vanish, and

$$\big[J_x, J_y\big] = \Big[J_x^{(1)}, J_y^{(1)}\Big] \otimes \mathbb{1} + \mathbb{1} \otimes \Big[J_x^{(2)}, J_y^{(2)}\Big] = \mathrm{i}\hbar J_z^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes \mathrm{i}\hbar J_z^{(2)} = \mathrm{i}\hbar J_z,$$

and the other two relations follow in the same way. The rotation formula is (10.2) of Proposition 10.2, with θ in place of t and the matrices $\mathbf{n} \cdot \mathbf{J}^{(1)}$, $\mathbf{n} \cdot \mathbf{J}^{(2)}$ in place of H_1, H_2. ∎

So the rotation of the pair by the angle θ about $\mathbf{n}$ turns each part by the same rotation. We call $\mathbf{J}$ the *total angular momentum*. By Theorem 7.3, the pair is an orthogonal sum of multiplets of $\mathbf{J}$. The question is which.

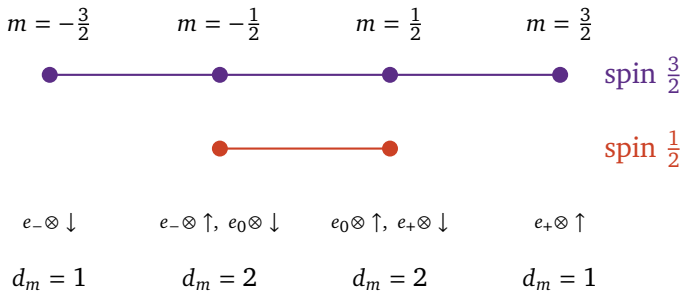


Figure 13.1: The weights of spin 1 times spin $\frac{1}{2}$. The six products (bottom) have $m = m_1 + m_2$, with the counts d_m. A multiplet of spin $\frac{3}{2}$ takes one vector at each m; the remaining vectors at $m = \pm\frac{1}{2}$ form a multiplet of spin $\frac{1}{2}$.

13.2 The multiplets of a product

Theorem 13.2 (Addition of angular momenta)**.** *Let $\mathbf{J}^{(1)}$ and $\mathbf{J}^{(2)}$ each be a single multiplet, of spins j_1 and j_2. Then $\mathbb{C}^{2j_1+1} \otimes \mathbb{C}^{2j_2+1}$ is an orthogonal sum of multiplets of $\mathbf{J}$, exactly one of each spin*

$$j = |j_1 - j_2|,\ |j_1 - j_2| + 1,\ \ldots,\ j_1 + j_2,$$

and of no other spin.

Proof. First, the products $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$ of the standard bases form an orthonormal basis of the pair, and by *Linear Algebra*, Prop. 12.9,

$$J_z\big(|j_1, m_1\rangle \otimes |j_2, m_2\rangle\big) = \hbar(m_1 + m_2)\,|j_1, m_1\rangle \otimes |j_2, m_2\rangle\,.$$

So the eigenspace of J_z for $\hbar m$ has the dimension d_m, the number of pairs (m_1, m_2) with $m_1 + m_2 = m$. Every such m differs from $j_1 + j_2$ by an integer, and $d_m = 0$ for every other m.

Next, count. The counts are symmetric, $d_{-m} = d_m$, so take $m \geq 0$. Let $j_1 \geq j_2$. The other case is the same with the two exchanged. Then $m_2 = m - m_1$ must lie between $-j_2$ and j_2, so m_1 runs over the values from $\max(-j_1,\ m - j_2)$ to $\min(j_1,\ m + j_2)$ in steps of 1. For $j_1 - j_2 \leq m \leq j_1 + j_2$ these are the values from $m - j_2$ to j_1, and for $0 \leq m \leq j_1 - j_2$ the values from $m - j_2$ to $m + j_2$. Hence

$$d_m = \begin{cases} j_1 + j_2 - m + 1, & j_1 - j_2 \leq m \leq j_1 + j_2, \\ 2j_2 + 1, & 0 \leq m \leq j_1 - j_2. \end{cases}$$

For j that differ from $j_1 + j_2$ by an integer, with $j_1 - j_2 \leq j \leq j_1 + j_2$, this gives $d_j - d_{j+1} = (j_1 + j_2 - j + 1) - (j_1 + j_2 - j) = 1$, also at $j = j_1 + j_2$, where $d_{j+1} = 0$. For $0 \leq j < j_1 - j_2$ both counts are $2j_2 + 1$, and $d_j - d_{j+1} = 0$.

Finally, by Theorem 7.3 the number of multiplets of spin j is $d_j - d_{j+1}$. So there is one multiplet of each spin from $j_1 - j_2$ to $j_1 + j_2$, and none of any other spin. ∎

As a check, the dimensions add up: $\sum_{j=|j_1-j_2|}^{j_1+j_2}(2j+1) = (2j_1+1)(2j_2+1)$. Each multiplet of $\mathbf{J}$ has its standard basis $|j, m\rangle$, the *coupled basis* of the pair. Its vectors are combinations of the products, and their coefficients are called *Clebsch–Gordan coefficients*. Within one multiplet the phases are fixed by Theorem 7.3, from the vector $|j, j\rangle$ down. The phase of $|j, j\rangle$ itself is a further convention, also due to Condon and Shortley: its coefficient on the product $|j_1, j_1\rangle \otimes |j_2, j - j_1\rangle$ is positive.

13.3 Two spins one half

Example 13.3 (Two spins one half)**.** For two spins one half, $S^{(1)}$ and $S^{(2)}$, we write $|{\uparrow\downarrow}\rangle = |{\uparrow}\rangle \otimes |{\downarrow}\rangle$, and so on, as in Lecture 10, where $|{\uparrow}\rangle = |0\rangle$.

1. *Which spins.* The weights are $m = 1, 0, 0, -1$, so by Theorem 13.2 the pair has spin 1 and spin 0.
2. *Spin* 1. The top vector is $|1,1\rangle = |{\uparrow\uparrow}\rangle$, the only product with $m = 1$. Since $S_- |{\uparrow}\rangle = \hbar\,|{\downarrow}\rangle$,

$$J_- |{\uparrow\uparrow}\rangle = \hbar\big(|{\downarrow\uparrow}\rangle + |{\uparrow\downarrow}\rangle\big) = \hbar\sqrt{2}\,|1,0\rangle\,,$$

   so $|1,0\rangle = \frac{1}{\sqrt{2}}(|{\uparrow\downarrow}\rangle + |{\downarrow\uparrow}\rangle)$, and lowering once more gives $|1,-1\rangle = |{\downarrow\downarrow}\rangle$.
3. *Spin* 0. The vector $|0,0\rangle$ has $m = 0$ and is orthogonal to $|1,0\rangle$, so it is a multiple of $|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle$. The Condon–Shortley convention makes the coefficient of $|{\uparrow\downarrow}\rangle$ positive:

$$|0,0\rangle = \tfrac{1}{\sqrt{2}}\big(|{\uparrow\downarrow}\rangle - |{\downarrow\uparrow}\rangle\big) = \psi^-.$$

So the triplet of Lecture 10 is the multiplet of spin 1 of the total spin, and the singlet ψ^- is the multiplet of spin 0. The names count the states. The singlet is entangled (Theorem 10.4), and it is the state of definite total spin 0. A state of definite total spin need not be a product. Conversely, a product need not have a definite total spin. For example, $|{\uparrow\downarrow}\rangle = \frac{1}{\sqrt{2}}(|1,0\rangle + |0,0\rangle)$, and a reading of J^2 on it gives total spin 1 or 0, each with probability $\frac{1}{2}$.

13.4 Spin one and spin one half

Figure 13.1 shows the next case: a spin 1 with the basis $e_+ = |1,1\rangle$, $e_0 = |1,0\rangle$, $e_- = |1,-1\rangle$ of the three-level system of Lecture 7, together with a spin one half.

Example 13.4 (Spin one and spin one half)**.**

1. *Which spins.* The weights are $\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}$, so the pair has spin $\frac{3}{2}$ and spin $\frac{1}{2}$.
2. *The top.* The only product with $m = \frac{3}{2}$ is $\left|\frac{3}{2},\frac{3}{2}\right\rangle = e_+ \otimes |{\uparrow}\rangle$.
3. *One step down.* By (7.5), $J_-^{(1)} e_+ = \hbar\sqrt{2}\,e_0$ and $J_-^{(2)} |{\uparrow}\rangle = \hbar\,|{\downarrow}\rangle$, so

$$J_-\big(e_+ \otimes |{\uparrow}\rangle\big) = \hbar\sqrt{2}\,e_0 \otimes |{\uparrow}\rangle + \hbar\,e_+ \otimes |{\downarrow}\rangle\,.$$

   The same vector is $\hbar\sqrt{3}\left|\frac{3}{2},\frac{1}{2}\right\rangle$, since the lowering coefficient at $j = m = \frac{3}{2}$ is $\sqrt{(\frac{3}{2}+\frac{3}{2})(\frac{3}{2}-\frac{3}{2}+1)} = \sqrt{3}$. Hence

$$\left|\tfrac{3}{2},\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\,e_0 \otimes |{\uparrow}\rangle + \sqrt{\tfrac{1}{3}}\,e_+ \otimes |{\downarrow}\rangle\,.$$

4. *The spin one half.* The vector $\left|\frac{1}{2},\frac{1}{2}\right\rangle$ has $m = \frac{1}{2}$ and is orthogonal to $\left|\frac{3}{2},\frac{1}{2}\right\rangle$. With the Condon–Shortley sign, positive on $e_+ \otimes |{\downarrow}\rangle$,

$$\left|\tfrac{1}{2},\tfrac{1}{2}\right\rangle = \sqrt{\tfrac{2}{3}}\,e_+ \otimes |{\downarrow}\rangle - \sqrt{\tfrac{1}{3}}\,e_0 \otimes |{\uparrow}\rangle\,.$$

   As a check, $J_+(e_+ \otimes |{\downarrow}\rangle) = \hbar\,e_+ \otimes |{\uparrow}\rangle$ and $J_+(e_0 \otimes |{\uparrow}\rangle) = \hbar\sqrt{2}\,e_+ \otimes |{\uparrow}\rangle$, so $J_+\left|\frac{1}{2},\frac{1}{2}\right\rangle = \hbar\big(\sqrt{\frac{2}{3}} - \sqrt{\frac{1}{3}}\sqrt{2}\big)e_+ \otimes |{\uparrow}\rangle = 0$, as it must, since $\left|\frac{1}{2},\frac{1}{2}\right\rangle$ is the top of its multiplet.

The vectors with negative m follow by lowering (Exercise Sheet 13).

The product $e_0 \otimes |\uparrow\rangle$ is $\sqrt{\frac{2}{3}}\left|\frac{3}{2},\frac{1}{2}\right\rangle - \sqrt{\frac{1}{3}}\left|\frac{1}{2},\frac{1}{2}\right\rangle$. A reading of J^2 on it gives the total spin $\frac{3}{2}$ with probability $\frac{2}{3}$ and $\frac{1}{2}$ with probability $\frac{1}{3}$.

The phases in the coupled basis are a convention. With the opposite sign of $\left|\frac{1}{2},\frac{1}{2}\right\rangle$ the basis would be another orthonormal basis of the same multiplets, and every probability would be the same.

13.5 Couplings

Corollary 13.5 (A coupling of two angular momenta). *Let $\mathbf{J}^{(1)}$ and $\mathbf{J}^{(2)}$ be single multiplets of spins j_1 and j_2, and let a be real. The matrix*

$$H = a\,\mathbf{J}^{(1)}\cdot\mathbf{J}^{(2)}, \qquad \mathbf{J}^{(1)}\cdot\mathbf{J}^{(2)} = \sum_{b=x,y,z} J_b^{(1)} \otimes J_b^{(2)},$$

commutes with J_x, J_y, J_z, and on the multiplet of total spin j it is the number

$$\frac{a\hbar^2}{2}\big(j(j+1) - j_1(j_1+1) - j_2(j_2+1)\big) \tag{13.2}$$

times $\mathbb{1}$. For $a \neq 0$ these numbers are distinct, so the levels of H are labelled by j, each of degeneracy $2j+1$.

Proof. Squaring each J_b of (13.1) and adding over b,

$$J^2 = J^{(1)2} \otimes \mathbb{1} + \mathbb{1} \otimes J^{(2)2} + 2\,\mathbf{J}^{(1)}\cdot\mathbf{J}^{(2)},$$

since $J_b^{(1)} \otimes \mathbb{1}$ and $\mathbb{1} \otimes J_b^{(2)}$ commute. On single multiplets $J^{(1)2} = \hbar^2 j_1(j_1+1)\mathbb{1}$ and $J^{(2)2} = \hbar^2 j_2(j_2+1)\mathbb{1}$ (Theorem 7.3). So $H = \frac{a}{2}\big(J^2 - \hbar^2 j_1(j_1+1) - \hbar^2 j_2(j_2+1)\big)$, a polynomial in J^2. It commutes with every J_a, since J^2 does (Lemma 7.2), and on the multiplet of spin j, where $J^2 = \hbar^2 j(j+1)\mathbb{1}$, it is (13.2). The number $j(j+1)$ grows with j, so for $a \neq 0$ the values differ. ∎

For $a = 0$ there is one level, the whole pair.

Hydrogen again. With $\mathbf{S} = \frac{\hbar}{2}\boldsymbol{\sigma}$ for each spin, $\mathbf{S}^{(1)}\cdot\mathbf{S}^{(2)} = \frac{\hbar^2}{4}\boldsymbol{\sigma}\cdot\boldsymbol{\sigma}$, so the Hamiltonian of Lecture 10 is $H = (\nu/\hbar)\,\mathbf{S}^{(1)}\cdot\mathbf{S}^{(2)}$. By (13.2) with $j_1 = j_2 = \frac{1}{2}$, it is $\frac{\hbar\nu}{2}(1\cdot 2 - \frac{3}{2}) = \frac{\hbar\nu}{4}$ on total spin 1 and $\frac{\hbar\nu}{2}(0 - \frac{3}{2}) = -\frac{3\hbar\nu}{4}$ on total spin 0. These are the levels of Example 10.6, now read off from the total spin.

Spin one and spin one half. For $j_1 = 1$, $j_2 = \frac{1}{2}$, (13.2) gives $\frac{a\hbar^2}{2}$ on $j = \frac{3}{2}$ and $-a\hbar^2$ on $j = \frac{1}{2}$. For $a \neq 0$ these are two levels, of degeneracies 4 and 2, and $E_{3/2} - E_{1/2} = \frac{3}{2}a\hbar^2$.

The coupling commutes with $\mathbf{J}$. For two spins one half it does not commute with the z component $S_z^{(1)} \otimes \mathbb{1}$ of one spin, if $a \neq 0$, since it then has the same eigenspaces as J^2, so by Lemma 4.3 a matrix commutes with the one exactly when it commutes with the other. And $S_z^{(1)} \otimes \mathbb{1}$ does not commute with J^2: $(S_z^{(1)} \otimes \mathbb{1})\psi^- = \frac{\hbar}{2}\cdot\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$, which is not a multiple of ψ^-. A matrix that commuted with J^2 would map the eigenspace of J^2 for 0, the line of ψ^-, into itself (Lemma 4.3). So for two spins one half the total spin and the z component of one spin are incompatible readings (Theorem 5.2). The coupled basis belongs to the complete set $\{J^2, J_z\}$ and the product basis to the complete set $\{S_z^{(1)} \otimes \mathbb{1},\ \mathbb{1} \otimes S_z^{(2)}\}$. Finally, "each spin once" is a statement about two single multiplets. Three spins one half give spin $\frac{1}{2}$ twice (Exercise Sheet 13).

13.6 Vector operators and the Zeeman effect

In its first excited level, $3p$, the outer electron of sodium has an orbital angular momentum $\mathbf{L}$ of spin 1 in the sense of Lecture 7 (the next course explains its origin) and its spin $\mathbf{S}$. This is the pair of Example 13.4. The coupling $\xi\,\mathbf{L}\cdot\mathbf{S}$, $\xi > 0$, of the spin to the orbital motion, the *spin–orbit coupling*, splits the level, by Corollary 13.5, into $j = \frac{3}{2}$ above $j = \frac{1}{2}$, $\Delta E/(hc) = 17.196\ \mathrm{cm}^{-1}$ (about 2.13 meV) apart. This splitting is the *fine structure* of the level. The lowest level, $3s$, has no orbital angular momentum. The labels $3p$ and $3s$ are the usual names of these sets of states, and the next course explains them. A magnetic field splits each level further, by an amount that depends on how $\mathbf{S}$ acts inside one multiplet of $\mathbf{J}$.

Definition 13.6 (Vector operator)**.** A triple $\mathbf{A} = (A_x, A_y, A_z)$ of matrices is a *vector operator* for an angular momentum $\mathbf{J}$ if

$$[J_a, A_b] = \mathrm{i}\hbar \sum_c \epsilon_{abc} A_c \quad \text{for all } a, b,$$

as $\mathbf{J}$ itself satisfies, with the symbol ϵ_{abc} of Lecture 1. Write $A_\pm = A_x \pm \mathrm{i}A_y$.

For the total $\mathbf{J}$ of this lecture, $\mathbf{J}^{(1)}\otimes\mathbb{1}$ and $\mathbb{1}\otimes\mathbf{J}^{(2)}$ are vector operators, since $[J_a, \mathbb{1}\otimes J_b^{(2)}] = \mathbb{1}\otimes[J_a^{(2)}, J_b^{(2)}]$.

Proposition 13.7 (Selection rule for m)**.** *For a vector operator, $[J_z, A_z] = 0$ and $[J_z, A_\pm] = \pm\hbar A_\pm$. So if $J_z v = \hbar m v$, then $A_z v$ is 0 or an eigenvector of J_z for $\hbar m$, and $A_\pm v$ is 0 or an eigenvector for $\hbar(m \pm 1)$. So a vector operator changes m by 0 or ± 1.*

Proof. $[J_z, A_z] = 0$ is the case $a = b$, and $[J_z, A_\pm] = \mathrm{i}\hbar A_y \pm \mathrm{i}(-\mathrm{i}\hbar A_x) = \pm\hbar A_\pm$. So $J_z A_\pm v = A_\pm J_z v \pm \hbar A_\pm v = \hbar(m \pm 1)A_\pm v$, and $J_z A_z v = \hbar m A_z v$. ∎

The rule is necessary, not sufficient. A change of m that it allows may still have a zero matrix element.

Inside one multiplet a vector operator has very little freedom. Let E be the projection onto the multiplet. By Proposition 13.7, EA_+E takes $|j, m\rangle$ to a multiple of $|j, m+1\rangle$, since each m occurs once. So EA_+E can differ from EJ_+E only by one factor at each step. The commutators force these factors to be equal (Figure 13.2).

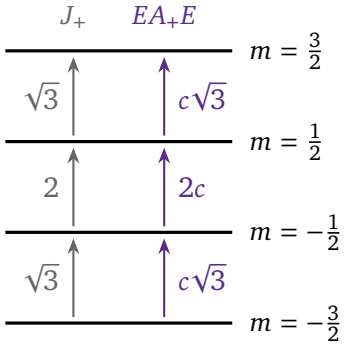


Figure 13.2: One multiplet, here of spin $\frac{3}{2}$, with coefficients in units of $\hbar$. Each m occurs once, so the matrices J_+ and EA_+E both move one rung up. The projection theorem says that the factor between them is the same number c on every rung.

Theorem 13.8 (Projection theorem). *Let E be the projection onto one multiplet, of spin j, in an orthogonal decomposition of the space into multiplets of $\mathbf{J}$. For every vector operator $\mathbf{A}$ there is a number c with $EA_aE = c\,EJ_aE$ for $a = x, y, z$. For $j = 0$ both sides vanish. For $j > 0$, $c = \operatorname{Tr}(E\,\mathbf{J}\cdot\mathbf{A})/\big(\hbar^2 j(j+1)(2j+1)\big)$.*

Proof. E commutes with every J_a, so the matrices $B_a = EA_aE$ form a vector operator too, acting on the multiplet. Expanding the definition gives $[J_+, B_+] = 0$, $[J_-, B_+] = -2\hbar B_z$ and $[J_-, B_z] = \hbar B_-$. In the multiplet each m occurs once, so by Proposition 13.7 $B_+\,|j,m\rangle$ is a multiple of $|j, m+1\rangle$. For $m < j$, write it as $c_m J_+\,|j,m\rangle$, which is possible since $J_+\,|j,m\rangle \neq 0$. Also $B_+\,|j,j\rangle = 0$. Applying $[J_+, B_+] = 0$ to $|j,m\rangle$ gives $c_m = c_{m+1}$ whenever $J_+^2\,|j,m\rangle \neq 0$, that is for $m \leq j-2$. So all c_m are one number c ($j = \frac{1}{2}$ has only one), and $B_+ = cJ_+$. Then $B_z = -[J_-, B_+]/(2\hbar) = -c[J_-, J_+]/(2\hbar) = cJ_z$ and $B_- = [J_-, B_z]/\hbar = cJ_-$. For $j = 0$, $B_+ = 0$ and the same two lines give $B_z = B_- = 0$. Finally, since E commutes with every J_a, $E\,\mathbf{J}\cdot\mathbf{A}\,E = c\,EJ^2E = c\hbar^2 j(j+1)E$, and its trace gives c. ∎

For $\mathbf{A} = \mathbb{1}\otimes\mathbf{J}^{(2)}$ and $j > 0$, $\mathbf{J}\cdot\mathbf{J}^{(2)} = \frac{1}{2}\big(J^2 + J^{(2)2} - J^{(1)2}\big)$ (as in Corollary 13.5), so $c = \big(j(j+1) + j_2(j_2+1) - j_1(j_1+1)\big)/\big(2j(j+1)\big)$.

Example 13.9 (The Landé factor). Take the $3p$ level as a model: the six states of $\mathbf{L}$ and $\mathbf{S}$, the nuclear spin and terms quadratic in the field neglected, and the spin's magnetic moment twice the orbital one per unit angular momentum. The electron's magnetic moment is opposite to its angular momentum, so a field B along z adds

$$H_1 = \frac{\mu_B B}{\hbar}\,(L_z + 2S_z) = \frac{\mu_B B}{\hbar}\,(J_z + S_z),$$

with the Bohr magneton μ_B, $\mu_B/h = 14.0$ GHz/T. On the $3p$ level $(L_z + 2S_z)/\hbar$ has the eigenvalues $2, 1, 0, 0, -1, -2$, so $\|H_1\| = 2\mu_B B$, a quarter of the fine-structure splitting or less for $B \leq 4.6$ T. So Theorem 6.8 applies to each of the two levels, with E the projection onto the level and g the fine-structure splitting. The first-order shifts are the eigenvalues of EH_1E, up to $3\,\|H_1\|^2/g$. By the projection theorem with $\mathbf{A} = \mathbf{S}$,

$$EH_1E = g_j\,\frac{\mu_B B}{\hbar}\,EJ_zE, \qquad g_j = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)},$$

where g_j is the *Landé factor*, and l and s are the spins of $\mathbf{L}$ and $\mathbf{S}$, here $l = 1$ and $s = \frac{1}{2}$. The factor is $\frac{4}{3}$ for $j = \frac{3}{2}$, $\frac{2}{3}$ for $j = \frac{1}{2}$, and 2 for the $3s$ level ($l = 0$). Each level of spin j splits into $2j+1$ equally spaced levels, shifted by $g_j m\,\mu_B B$, to within $3\,\|H_1\|^2$ divided by the fine-structure splitting. Because the g_j differ, the sodium lines split into more components than the three of a factor 1 in place of 2 (Exercise Sheet 13).

The projection theorem is about one multiplet. The spin $\mathbf{S}$ also connects the two multiplets of the $3p$ level, and the theorem says nothing about those matrix elements. Moreover, g_j belongs to a weak field, and Exercise Sheet 13 takes any field.

The physics

The yellow light of sodium lamps is two lines, at 589.0 and 589.6 nm in air. Both come from the outer electron of the sodium atom falling from its first excited level, $3p$, to its lowest level, $3s$ (§ 13.6). The line at 589.0 nm starts from $j = \frac{3}{2}$ and the line at 589.6 nm from $j = \frac{1}{2}$. Zeeman saw spectral lines split in a magnetic field in 1896. The sodium lines split into more components than the three of the "normal" effect,

and this "anomalous" Zeeman effect was understood only after the electron's spin was proposed, by Uhlenbeck and Goudsmit in 1925.

Adding angular momenta

Total: $J_a = J_a^{(1)} \otimes \mathbb{1} + \mathbb{1} \otimes J_a^{(2)}$ is an angular momentum (Lemma 13.1). *Addition:* spins j_1 and j_2 give one multiplet of each spin from $|j_1 - j_2|$ to $j_1 + j_2$ (Theorem 13.2). *Two spins one half:* triplet $|\uparrow\uparrow\rangle$, $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$, $|\downarrow\downarrow\rangle$ and singlet ψ^- (Example 13.3). *Coupling:* $a\,\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)}$ is $\frac{a\hbar^2}{2}\big(j(j+1) - j_1(j_1+1) - j_2(j_2+1)\big)$ on total spin j (Corollary 13.5). *Projection:* $EA_aE = c\,EJ_aE$ on a multiplet; Landé $g_j = 1 + \frac{j(j+1)+s(s+1)-l(l+1)}{2j(j+1)}$ (Theorem 13.8, Example 13.9).

Summary.

- *Addition of angular momenta*: two multiplets of spins j_1 and j_2 give one multiplet of each spin from $|j_1 - j_2|$ to $j_1 + j_2$ (Theorem 13.2).
- *Two spins one half*: the triplet and the singlet are the total spins 1 and 0 (Example 13.3).
- *Couplings*: for $a \neq 0$, $a\,\mathbf{J}^{(1)} \cdot \mathbf{J}^{(2)}$ has one level for each total spin (Corollary 13.5).
- *Projection*: inside one multiplet a vector operator is a multiple of $\mathbf{J}$; a weak field splits a level by $g_j m\,\mu_B B$ (Theorem 13.8, Example 13.9).

The triplet is symmetric under the swap of the two spins, and the singlet is antisymmetric. Lecture 14 shows why this matters when the two spins belong to two particles of the same kind.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §3.8, pp. 205–218. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 2, Ch. X, §B, pp. 1019–1024, §C, pp. 1025–1039, and Complement D_X, pp. 1065–1075. Sakurai and Napolitano, §3.11, pp. 231–240, and §5.3.3, pp. 311–314. NIST Atomic Spectra Database (sodium).